Joint Estimation of Target Dose and Slope in Binary Response Models: A Geometric Approach

Maider Sada[1*], José A. Moler[1] and Nancy Flournoy[2]

[1*]Department of Statistics, Informatics and Mathematics, Universidad Pública de Navarra, Campus Arrosadia s/n, Pamplona, 31006, Spain.
[2]Department of Statistics, University of Missouri, Columbia, Missouri, USA.

*Corresponding author(s). E-mail(s): maider.sada@unavarra.es;
Contributing authors: jmoler@unavarra.es; flournoyn@missouri.edu;

Abstract

A common goal in many fields is to identify the level of a stimulus (dose) that has a prescribed percentage of positive binary responses. We also consider the estimation of the slope of the response function at the dose of interest to calibrate the importance of error in the dose estimate. We use a parameterization whose information matrix depends directly on the parameters of interest. By mapping the problem to an induced dose space, the optimization process becomes independent of the unknown parameters. This dependency, however, is reintroduced when transforming the optimal doses from the induced space back into the real dose space.

Unfortunately, the symmetrical properties that appear in the geometrical reasoning of previous works no longer hold. In contrast, the geometry presented here provides a mechanistic understanding that overcomes that of two-dimensional graphic methods. For example, one can explicitly observe how designs are constructed from convex combinations of elementary designs, revealing the specific contribution of each dose to the total information and providing a comprehensive view of the matrices of all possible designs.

For a wide class of optimality criteria and response models, we find that the optimal design for the joint estimation of the dose and the slope is a two-point design symmetric about the median. This discovery reduces the computational complexity from a five-parameter to a single-parameter problem as we show for finding the standardized A- and cc-optimal designs introduced in Flournoy et al. (2025).

Keywords: dose-finding designs, binary response models, optimal designs, logistic, probit

MSC Classification: 62K05 , 60F10 , 62J12

1 Introduction

Dose-finding experiments are common in quite different fields. In clinical research, toxicological studies consider a dose as a concentration of the new drug where a positive response can represent toxicity or efficacy depending on the clinical trial's objective; see Oron et al. (2022). In sensory studies, doses are the different levels of a stimulus applied to individuals, see Leek (2001) or Wetherill and Levitt (1965). In engineering, doses can represent different levels of tension to measure materials' physical or energetic resistance or sensitivity; see Lagoda and Sonsino (2004), Pollak et al. (2006) and Christensen et al. (2024). In education, doses represent the degree of difficulty of test items; see Chang and Ying (2009). We assume that each observation is collected from a different individual, so that responses are independent conditional upon dose. We call the allowable doses the *dose space* $\mathcal{X}$, which we initially assume to be an unrestricted continuous space, that is, $\mathcal{X} = \mathbb{R}$, to explore global optimal designs. Furthermore, we assume that the probability of having a positive response increases with dose. In particular, we assume that the dose-response relationship can be mathematically expressed by the continuous and monotonically increasing function

$$F(x, \boldsymbol{\theta}) = P(\textit{Positive response} \mid \textit{Dose} = x), \tag{1}$$

where the vector of unknown parameters $\boldsymbol{\theta}$ is suppressed unless it is necessary for understanding. We refer to $F(x, \boldsymbol{\theta})$ as the model.

More formally, dose-finding experiments aim to estimate a dose μ_Γ that has a positive response rate Γ, that is, to find μ_Γ such that $F(\mu_\Gamma) = \Gamma$. In this context, it is also of interest to estimate $\eta_\Gamma = F'(\mu_\Gamma)$, the slope of the dose-response function at the μ_Γ dose. This slope is relevant because, when it is large, even a small error in the estimation of μ_Γ will result in a dose estimate whose positive response rate is far from Γ; while when the slope is small, a large error in the estimate of μ_Γ conveys negligible changes in its associated positive response rate.

Our starting point is the standard binary magentalocation-scale model, where the probability of a positive response at dose x is given by $F(x) = W[(x-\mu)\eta]$. Here, $W(\cdot)$ denotes the logistic or probit function, while μ and $\eta > 0$ represent the location and scale parameters, respectively. However, our actual objective is the joint estimation of a specific target dose μ_Γ and the model sensitivity at that point, η_Γ. Although these can be expressed in terms of the original parameters μ and η as

$$\mu_\Gamma = \frac{W^{-1}(\Gamma)}{\eta} + \mu \quad \text{and} \quad \eta_\Gamma = W'(\Gamma)\eta,$$

it is preferable to define a model that takes the values to be estimated as its direct parameters. By doing so, any criterion aimed at minimizing the variance of the model parameters directly minimizes the values of interest. Thus, we define the model through a function H such that

$$H[(x - \mu_\Gamma)\eta_\Gamma] = F(x). \tag{2}$$

Once the model is parameterized in terms of the target parameters, $(\mu_\Gamma, \eta_\Gamma)$ can be estimated from experimental data using the Maximum Likelihood (ML) method. Suppose that experiment is conducted that produces a sample of n independent binary responses $y_j \in \{0, 1\}$ at doses x_j, for $j = 1, \ldots, n$. The log-likelihood function under the reparameterized model is given by

$$\ell(\mu_\Gamma, \eta_\Gamma) = \sum_{j=1}^{n} \left[y_j \ln(H[(x_j - \mu_\Gamma)\eta_\Gamma]) + (1 - y_j) \ln(1 - H[(x_j - \mu_\Gamma)\eta_\Gamma]) \right].$$

Maximum likelihood estimators (MLEs), denoted by $(\hat{\mu}_\Gamma, \hat{\eta}_\Gamma)$, are the values that maximize this function. Given the non-linear nature of H, these estimators are typically computed using standard numerical optimization algorithms. The asymptotic precision of these estimators depends on x_j, the experimental doses chosen. This dependence naturally motivates the main objective of this paper: finding optimal designs that maximize the information gathered about $(\mu_\Gamma, \eta_\Gamma)$.

Ford et al. (1992) obtain explicit D-optimal designs for (1) with a parameterization that differs from (2) for a wide catalog of models. As D-optimal designs are invariant under linear transformations of the parameters [see Fedorov (1972) p.81], they are also D-optimal under (2). Only for $\Gamma = 0.5$ using parameterization (2), Sitter and Fainaru (1997) establish a geometrical procedure for obtaining optimal designs for a set of optimality criteria and models that satisfy some regularity conditions. These regularity conditions are fulfilled in usual practical situations, for instance, for the logistic and probit models and D-, A- and E- optimality criteria. magenta In Fedorov and Leonov (2014) the treatment of several optimality criteria is deeply reviewed for binary response models. magentaHowever, since common optimal designs for nonlinear response models are functions of the unknown parameters, they cannot be directly implemented. Rather than worry about the robustness of procedures based on parameter guesses, we recommend using adaptive designs that have been shown to converge to optimal designs. In addition to improving efficiency, adaptive designs provide observations at more points than the theoretical optimal design prescribes, which can be useful for diagnostics and other model checking. Fedorov and Leonov (2014) also describe how adaptive designs can be used to implement optimal designs in our context.

Finally, with parameterization (2), Flournoy et al. (2025) obtain, under different models, explicit mathematical expressions for the c-optimal designs and the variances of the estimates. Then they introduce the cc-optimality criterion that minimizes the variance of $\widehat{\eta}_\Gamma$ subject to an imposed maximum variance of $\hat{\mu}_\Gamma$, with each variance standardized with respect to the variance from their respective c-optimal designs. Then, writing the cc-optimal design as a compound design [Fedorov and Gaivoronski (1984)] they prove it has two or three support points depending on the model and propose an algorithm that makes an exhaustive search that evaluates the cc-optimality criterion for the four parameters in the two-doses designs and the six in the three-dose designs.

In this work, we extend prior results in three different ways.

- The geometrical procedure in Sitter and Fainaru (1997) is extended to any value $\Gamma \in (0, 1)$ and, in this new framework, we establish optimal designs that satisfy Sitter and Fainaru's conditions.
- A more efficient procedure is given for obtaining the cc-optimal design proposed by Flournoy et al. (2025) in the unrestricted dose space and a procedure for restricted dose spaces is proposed.
- Mathematically explicit expressions are given for the standardized A-optimal designs proposed by Dette (2002).

This work is organized as follows. The reparameterization $(x - \mu_\Gamma)\eta_\Gamma$ is described in Section 2. Theorem 3 in Section 3 makes computation more efficient by generalizing Sitter and Fainaru (1997) geometry. In Section 4, A and cc-optimal designs are defined as single-parameter optimization problems, which provides an efficient computational procedure for unrestricted and restricted dose spaces. Section 5 presents a comparative numerical study assessing the performance of the designs proposed in Section 4. The paper closes with a discussion in Section 6.

2 The $(x - \mu_\Gamma)\eta_\Gamma$ Parameterization

Sitter and Fainaru (1997) provide conditions under which the optimal design, for a known class of criteria and $\Gamma = 0.5$, consists of two doses symmetric about the median. Their results are illustrated for the logistic and probit models (see Table 1). Applying the same geometric arguments, we will show that the optimal design for any $\Gamma \in (0, 1)$ also comprises the median and two symmetric points relative to the median.

Model	$W(s)$	$W'(s)$	$w^2(s)$
Logistic	$[1 + \exp(-s)]^{-1}$	$W[1 - W](s)$	$W[1 - W](s)$
Probit	$\int_{-\infty}^{s} f(t)dt$	$f(s) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{s^2}{2}\right)$	$\frac{f^2}{W[1 - W]}(s)$

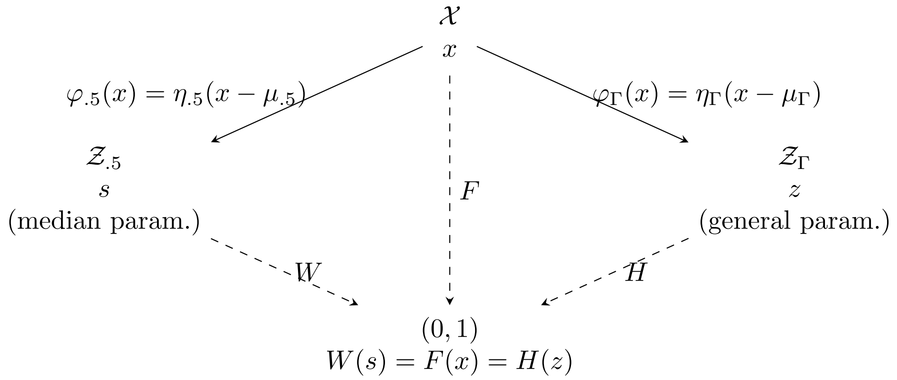
Table 1 Toxicity models under initial parameterization $(x - \mu_{.5})\eta_{.5}$.

To extend this work to any value of $\Gamma \in (0, 1)$, the first step is to redefine the parameterization by explicitly including the parameters of interest μ_Γ and η_Γ. Sitter and Fainaru (1997) write $F(x)$ as

$$F(x) = W[(x - \mu_{.5})\eta_{.5}],$$

where W is the logistic or probit model with unknown parameters $\mu_{.5} = F^{-1}(0.5)$ and $\eta_{.5} = F'(\mu_{.5})$. To generalize this expression to $\Gamma \in (0,1)$, we need to define a suitable function H such that $H[(x-\mu_\Gamma)\eta_\Gamma] = F(x) = W[(x-\mu_{.5})\eta_{.5}]$. Then, H must satisfy $\Gamma = F(\mu_\Gamma) = H(0)$ and $\eta_\Gamma = F'(\mu_\Gamma) = H'(0)\eta_\Gamma$ so $H'(0) = 1$. This suitable function H can be defined by the scheme in Figure 1.

Fig. 1 Scheme of parameterizations from the real dose space $\mathcal{X}$ through the induced dose space $\mathcal{Z}_{.5}$ to $\mathcal{Z}_\Gamma$, $\Gamma \neq 0.5$.



First look down the center of Figure 1 where the *real dose space* $\mathcal{X}$ and the unknown *toxicity function* F are shown. Under $\mathcal{X}$ we define two *induced dose spaces* $\mathcal{Z}_{.5}$ and $\mathcal{Z}_\Gamma$, for any $\Gamma \neq 0.5$, by the transformations $\varphi_{.5}(x) = (x-\mu_{.5})\eta_{.5} = s$ and $\varphi_\Gamma(x) = (x-\mu_\Gamma)\eta_\Gamma = z$.

To satisfy $H(z) = F(x) = W(s)$,

$$H(z) = F\left(\frac{z}{\eta_\Gamma} + \mu_\Gamma\right) = W\left(z\frac{\eta_{.5}}{\eta_\Gamma} + (\mu_\Gamma - \mu_{.5})\eta_{.5}\right).$$

Defining $\nu = (\mu_\Gamma - \mu_{.5})\eta_{.5}$ and $\zeta = \dfrac{\eta_\Gamma}{\eta_{.5}}$, then

$$H(z) = W(s) \text{ with } s = \frac{z}{\zeta} + \nu. \tag{3}$$

On the other hand, if we set $H(0) = \Gamma$ and $H'(0) = 1$ as we desire, then

$$\Gamma = H(0) = W(\nu) \quad \text{and} \quad 1 = H'(0) = \frac{W'(\nu)}{\zeta}.$$

So, defining $\nu = W^{-1}(\Gamma)$ and $\zeta = W'(\nu)$, which depend only on the known function W and Γ value, $H(z)$ can be defined as in (3) without dependence on unknown parameters μ_Γ and η_Γ. This allows us to define both the function H and the optimal design in $\mathcal{Z}_\Gamma$ without external information.

3 Characterization of the Optimal Designs

In this section, we generalize the work of Sitter and Fainaru (1997). We begin by defining the space of information matrices and adapt it to the parameters $(\mu_\Gamma, \eta_\Gamma)$. Then we extend Sitter and Fainaru's definition of Vertical Boundary ($\mathcal{VB}$) and, finally, we generalize their main result.

3.1 Space of the information matrix

Let Ξ_Γ be the set of all possible designs with the doses in $\mathcal{Z}_\Gamma = \varphi_\Gamma(\mathcal{X})$, and define a design $\xi \in \Xi_\Gamma$ as

$$\xi = \begin{Bmatrix} z_1 & \dots & z_k \\ \omega_1 & \dots & \omega_k \end{Bmatrix}, \tag{4}$$

where w_i represents the proportion of patients assigned to dose z_i. For a binary response model consider $P(y = 1) = H(z)$. Denoting the transpose with the superscript t, the score function becomes $\{H'(z)(y - H(z))/[H(z)(1 - H(z))]\}(-\eta_\Gamma,\ z/\eta_\Gamma)^t$. Therefore the information matrix of a design ξ is

$$\mathrm{I}(\xi) = \sum_{i=1}^{k} \omega_i h^2(z_i) \begin{pmatrix} \eta_\Gamma^2 & -z_i \\ -z_i & \frac{z_i^2}{\eta_\Gamma^2} \end{pmatrix} = \begin{pmatrix} t_\Gamma(\xi) & u_\Gamma(\xi) \\ u_\Gamma(\xi) & v_\Gamma(\xi) \end{pmatrix}, \tag{5}$$

where

$$t_\Gamma(\xi) = \eta_\Gamma^2 \sum_{i=1}^{k} \omega_i\ h^2(z_i),$$
$$u_\Gamma(\xi) = -\sum_{i=1}^{k} \omega_i\ h^2(z_i)\ z_i,$$
$$v_\Gamma(\xi) = \frac{1}{\eta_\Gamma^2} \sum_{i=1}^{k} \omega_i\ h^2(z_i)\ z_i^2;$$

and

$$h^2(z) = \frac{(H')^2}{H(1-H)}(z) = \frac{1}{\zeta^2}\frac{(W')^2}{W(1-W)}(s) = \frac{1}{\zeta^2} w^2(s), \tag{6}$$

for $w^2(s)$ defined in Table 1. Equation (5) is numerically equivalent to expression (1.1) in Biedermann et al. (2006) but, it is not conceptually the same. Following Sitter and Fainaru (1997), the information matrix of every design can be expressed using a vector $(t_\Gamma, u_\Gamma, v_\Gamma)(\xi)$ in $\mathbb{R}^3$. This allows us to work with $\mathbb{R}^3$ instead of Ξ_Γ, which makes it easier to interpret and characterize the optimal design.

Let $\mathcal{C}_\Gamma : \Xi_\Gamma \to \mathbb{R}^3$ where $\mathcal{C}_\Gamma(\xi) = (t_\Gamma, u_\Gamma, v_\Gamma)(\xi)$. Then, the optimal design can be found maximizing a known function $\phi : \mathbb{R}^3 \to \mathbb{R}$ in the set $\mathcal{C}_\Gamma(\Xi_\Gamma)$. In order to define $\mathcal{C}_\Gamma(\Xi_\Gamma)$, we first consider single-point designs. Let $z \in \mathcal{Z}_\Gamma$, and ξ_z be the single-point design defined by z. In this case, by (5),

$$\mathcal{C}_\Gamma(\xi_z) = h^2(z) \begin{pmatrix} \eta_\Gamma^2 \\ -z \\ \dfrac{z^2}{\eta_\Gamma^2} \end{pmatrix}. \tag{7}$$

Observe that this expression depends only on $z \in \mathcal{Z}_\Gamma$, so following Sitter and Fainaru (1997) we can define the *space curve* $C_\Gamma : \mathcal{Z}_\Gamma \to \mathbb{R}^3$ where $C_\Gamma(z) = \mathcal{C}_\Gamma(\xi_z)$, which represents the set of information matrices for all single-dose designs. The space curve $C_\Gamma(\mathcal{Z}_\Gamma)$ is shown in Figure 2 for the logistic function and different values of Γ.

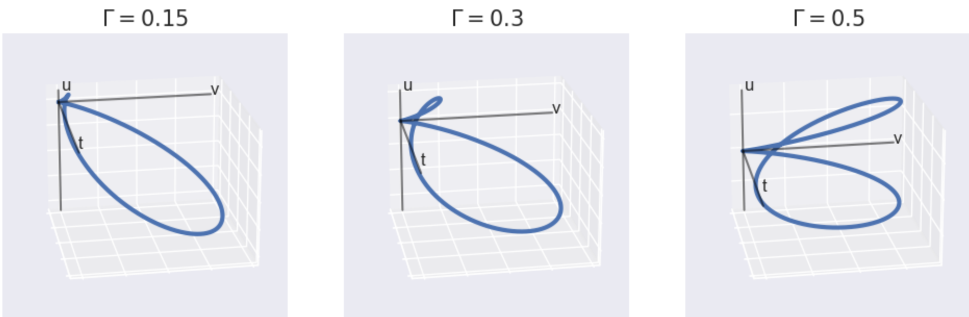


Fig. 2 $C_\Gamma(\mathcal{Z}_\Gamma)$ for the logistic model and $\Gamma = 0.15$, 0.3 and 0.5.

This space curve will be the basis for building $\mathcal{C}_\Gamma(\Xi_\Gamma)$. For any design $\xi \in \Xi_\Gamma$ defined in (4), it follows from expression (6) and the definition of $C_\Gamma(z)$ that

$$\mathcal{C}_\Gamma(\xi) = \sum_{i=1}^{k} \omega_i C_\Gamma(z_i).$$

In words, $\mathcal{C}_\Gamma(\xi)$ represents a convex combination of k points on the curve $C_\Gamma(\mathcal{Z}_\Gamma)$. Therefore, the set $\mathcal{C}_\Gamma(\Xi_\Gamma)$ constitutes the convex hull of $C_\Gamma(\mathcal{Z}_\Gamma)$, defined as the set of all such convex combinations and denoted in what follows as $\mathcal{CH}_\Gamma$.

Figure 3 illustrates the construction of $\mathcal{CH}_\Gamma$ for $\Gamma = 0.3$. Figure 3 a) is Figure 2 b). Figure 3 (b) geometrically represents a two-point design $\begin{Bmatrix} z_1 & z_2 \\ \omega & 1-\omega \end{Bmatrix}$ between the points corresponding to the single-point designs $z_1, z_2 \in \mathcal{Z}_{0.3}$. Figure 3 (c) shows the convex hull $\mathcal{CH}_{0.3}$.

Using this geometric approach, and applying Caratheodory's theorem [Appendix 2 from Silvey (1980)], we have that any possible information matrix can be computed

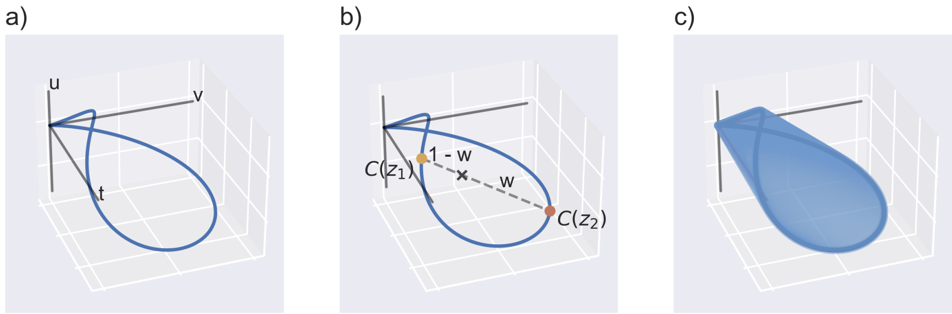


Fig. 3 Construction of $\mathcal{CH}_{0.3}$.

using at most $\dim(\mathbb{R}^3)+1 = 4$ points of $C_\Gamma(\mathcal{Z}_\Gamma)$ and, consequently, a four point design. We also have that, for the usual criteria (D, A, E, ...), the optimal designs' information matrix is in the boundary of $\mathcal{CH}_\Gamma$, which is defined using at most $\dim(\mathbb{R}^3) = 3$ points.

3.2 The Right Boundary

Sitter and Fainaru (1997) use their geometric approach to define the *Vertical Boundary* ($\mathcal{VB}$), a set that contains the optimal design for any criterion Φ that satisfies the following condition:

Condition 1. $\Phi[I(\xi)] = \phi[t_\Gamma(\xi), u_\Gamma(\xi), v_\Gamma(\xi)]$ *is nondecreasing with respect to* $v_\Gamma(\xi)$.

The following definition generalizes Sitter and Fainaru (1997)'s Vertical Boundary for $\Gamma \in (0, 1)$.

Definition 1. *Let* $\mathcal{Z}_\Gamma$ *and* $\mathcal{CH}_\Gamma$ *be as previously defined. Then the Right Boundary of* $\mathcal{CH}_\Gamma$ *is*

$$\mathcal{RB}_\Gamma = \bigcup_{\substack{t \in t_\Gamma(\Xi_\Gamma) \\ u \in u_\Gamma(\Xi_\Gamma)}} \operatorname*{argmax}_{(t,u,v) \in \mathcal{CH}_\Gamma} \{v\}. \tag{8}$$

When $\Gamma = 0.5$, $\mathcal{RB}_{.5} = \mathcal{VB}$, and when $\Gamma \neq 0.5$, the set plays an equivalent role. That is, for a criterion that satisfies Condition 1, the optimal design lies in $\mathcal{RB}_\Gamma$. A detailed construction of this set, along with the justification that it contains the optimal design, is given in Appendix A. Right bounds,$\mathcal{RB}_\Gamma$, for the curves in Figure 2 ared shaded in Figure 4.

Sitter and Fainaru (1997) give the following conditions for W in order to easily characterize $\mathcal{VB}$:

Condition 2. $W(-s) = 1 - W(s)$.

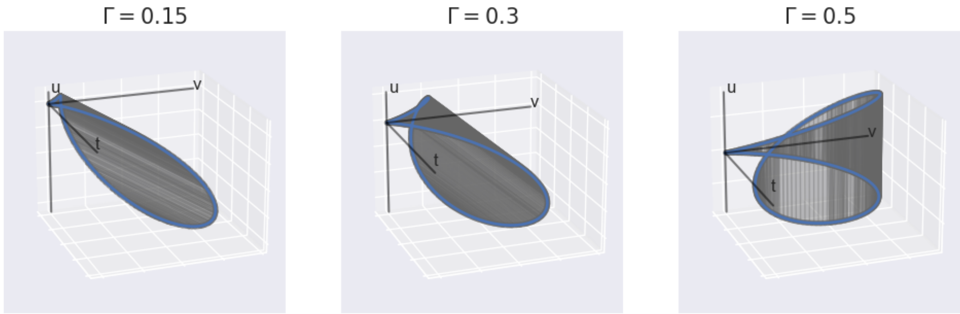


Fig. 4 The Right Boundary ($\mathcal{RB}_\Gamma$) for the logistic model and $\Gamma = 0.1$, 0.3 and 0.5.

Condition 3. *The curve* $\{w^2(s)(1, s) \mid s \in \mathcal{Z}_{.5}\}$ *is concave in* $\mathbb{R}^2$.

Condition 4. *The curve* $\{w^2(s)(1, s^2) \mid s \in \mathcal{Z}_{.5}\}$ *is concave in* $\mathbb{R}^2$.

Under these conditions, Sitter and Fainaru shown that the Vertical Boundary (i.e., the Right Boundary of $\mathcal{CH}_{.5}$) can be defined as

$$\mathcal{RB}_{.5} = \mathcal{VB} = \{\omega C_{.5}(-s) + (1-\omega)\, C_{.5}(s) \mid s \in \mathcal{Z}_{.5}, \omega \in [0,1]\}\,.$$

In words, the set $\mathcal{RB}_{.5}$ is the information matrix of two-point designs in which the doses are symmetric to the median.

Remark 1. *The logistic and probit models (Table 1) represent the standard models that satisfy conditions 2, 3, and 4. However, the generalization presented in this work can also be applied to double exponential and double reciprocal models. Although these models only satisfy conditions 2* and *3, Sitter and Wu (1993) showed that the optimal design, for a criteria that satisfies Condition 1, for* $\Gamma = 0.5$ *consists of three doses: the median and two points symmetric about the median. Applying the same geometric arguments developed in this paper, it follows, that, for the double exponential and double reciprocal models, the optimal design for any* $\Gamma \in (0,1)$ *also comprises the median and two symmetric points relative to the median.*

3.3 Generalization to $\Gamma \in (0, 1)$

We use the case $\Gamma = 0.5$ as the starting point for characterizing the generalized set $\mathcal{RB}_\Gamma$. The following proposition extends the relationship between both induced dose spaces $\mathcal{Z}_\Gamma$ and $\mathcal{Z}_{.5}$ (see Section 2) to the spaces of information matrices.

Proposition 1. *For $\Gamma \in (0,1)$, there exists a linear and bijective application L such that $C_\Gamma(z) = LC_{.5}(s)$. This application and its inverse are given by*

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \dfrac{\nu}{\zeta\eta_{.5}^2} & \dfrac{1}{\zeta} & 0 \\ \dfrac{\nu^2}{\zeta^2\eta_{.5}^4} & \dfrac{2\nu}{\zeta^2\eta_{.5}^2} & \dfrac{1}{\zeta^2} \end{pmatrix} \text{ and } L^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -\dfrac{\nu}{\eta_{.5}^2} & \zeta & 0 \\ \dfrac{\nu^2}{\eta_{.5}^4} & -2\nu\dfrac{\zeta}{\eta_{.5}^2} & \zeta^2 \end{pmatrix}.$$

Proof Using (7) and $h^2(z) = w^2(s)/\zeta^2$,

$$C_\Gamma(z) = h^2(z)\begin{pmatrix} \eta_\Gamma^2 \\ -z \\ \dfrac{z^2}{\eta_\Gamma^2} \end{pmatrix} = \frac{w^2(s)}{\zeta^2}\begin{pmatrix} \eta_\Gamma^2 \\ -z \\ \dfrac{z^2}{\eta_\Gamma^2} \end{pmatrix}.$$

Using (3),

$$C_\Gamma(z) = \frac{w^2(s)}{\zeta^2}\begin{pmatrix} \eta_\Gamma^2 \\ -(s-\nu)\zeta \\ \dfrac{(s-\nu)^2\zeta^2}{\eta_\Gamma^2} \end{pmatrix}.$$

Using $\eta_\Gamma = \eta_{.5}\zeta$,

$$\begin{aligned} C_\Gamma(z) = w^2(s) &\begin{pmatrix} \eta_{.5}^2 \\ \dfrac{\nu}{\zeta} - \dfrac{s}{\zeta} \\ \dfrac{\nu^2}{\eta_{.5}^2\zeta^2} - 2\nu\dfrac{s}{\eta_{.5}^2\zeta^2} + \dfrac{s^2}{\eta_{.5}^2\zeta^2} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ \dfrac{\nu}{\zeta\eta_{.5}^2} & \dfrac{1}{\zeta} & 0 \\ \dfrac{\nu^2}{\zeta^2\eta_{.5}^4} & \dfrac{2\nu}{\zeta^2\eta_{.5}^2} & \dfrac{1}{\zeta^2} \end{pmatrix} w^2(s)\begin{pmatrix} \eta_{.5}^2 \\ -s \\ \dfrac{s^2}{\eta_{.5}^2} \end{pmatrix} = LC_{.5}(s); \end{aligned}$$

and thus

$$L^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -\dfrac{\nu}{\eta_{.5}^2} & \zeta & 0 \\ \dfrac{\nu^2}{\eta_{.5}^4} & -\dfrac{2\nu\zeta}{\eta_{.5}^2} & \zeta^2 \end{pmatrix}.$$

$\square$

Given $\Gamma \in (0,1)$ and the transformation L defined in Proposition 1, the bijectivity of L establishes a direct correspondence between the information matrices for the single-point designs in $\mathcal{Z}_\Gamma$ and $\mathcal{Z}_{.5}$. Moreover, since L is linear, it preserves convex combinations. Consequently, $\mathcal{CH}_\Gamma = L(\mathcal{CH}_{.5})$, and the correspondence between the information matrices is extended to all designs. Similarly, this linear bijectivity ensures the preservation of the topological boundaries, mapping the boundary of $\mathcal{CH}_{.5}$ to the boundary of $\mathcal{CH}_\Gamma$.

The next proposition extends the relationship between convex hulls in $\mathcal{Z}_\Gamma$ and $\mathcal{Z}_{.5}$ to their Right Boundaries, which makes it possible to compute $\mathcal{RB}_\Gamma$ (the set in which the optimal designs lie) from $\mathcal{RB}_{.5} = \mathcal{VB}$.

Proposition 2. *Let $\Gamma \in (0,1)$ and L be defined in Proposition 1 with $\mathcal{RB}_\Gamma$ and $\mathcal{RB}_{.5}$. Then,*

$$\mathcal{RB}_\Gamma = L\,\mathcal{RB}_{.5} = \left\{\omega C_\Gamma(-\nu\zeta - z) + (1-\omega)C_\Gamma(-\nu\zeta + z) \mid z \in \mathcal{Z}_\Gamma\right\}.$$

Proof From Proposition 1,

$$\begin{pmatrix} t_\Gamma \\ u_\Gamma \\ v_\Gamma \end{pmatrix} = L \begin{pmatrix} t_{.5} \\ u_{.5} \\ v_{.5} \end{pmatrix}.$$

Then, starting from Definition 1 of $\mathcal{RB}_\Gamma$,

$$\begin{aligned}
\mathcal{RB}_\Gamma &= \bigcup_{\substack{t \in t_\Gamma(\Xi_\Gamma) \\ u \in u_\Gamma(\Xi_\Gamma)}} \underset{(t,u,v)\in\mathcal{CH}_\Gamma}{\operatorname{argmax}} \{v\} \\
&= L\left(\bigcup_{\substack{t_0 \in t_{.5}(\Xi_{.5}) \\ u_0 \in u_{.5}(\Xi_{.5})}} \underset{(t_0,u_0,v_0)\in\mathcal{CH}_{.5}}{\operatorname{argmax}} \left\{\frac{\nu^2}{\zeta^2\eta_{.5}^4} t_0 + 2\frac{2\nu}{\zeta^2\eta_{.5}^2} u_0 + \frac{v_0}{\zeta^2}\right\}\right).
\end{aligned}$$

Since the values t_0, u_0 and ζ are constant within each set of the union, the maximization depends solely on v_0:

$$\mathcal{RB}_\Gamma = L\left(\bigcup_{\substack{t_0 \in t_{.5}(\Xi_{.5}) \\ u_0 \in u_{.5}(\Xi_{.5})}} \underset{(t_0,u_0,v_0)\in\mathcal{CH}_{.5}}{\operatorname{argmax}} \{v_0\}\right) = L\,\mathcal{VB}.$$

Finally, since L is a linear map, it preserves convex combinations. Thus,

$$\begin{aligned}
\mathcal{RB} = L\,\mathcal{VB} &= L\left\{\omega C_{.5}(-s) + (1-\omega)\, C_{.5}(s) \mid s \in \mathcal{Z}_{.5}\right\} \\
&= \left\{\omega L C_{.5}(-s) + (1-\omega)\, L C_{.5}(s) \mid s \in \mathcal{Z}_{.5}\right\} \\
&= \left\{\omega C_\Gamma(-\nu\zeta - s\zeta) + (1-\omega)\, C_\Gamma(-\nu\zeta + s\zeta) \mid s \in \mathcal{Z}_{.5}\right\} \\
&= \left\{\omega C_\Gamma(-\nu\zeta - z) + (1-\omega)\, C_\Gamma(-\nu\zeta + z) \mid z \in \mathcal{Z}_\Gamma\right\}.
\end{aligned}$$

□

The main idea behind Proposition 2 is that the set $\mathcal{RB}_\Gamma$ is defined by lines connecting pairs of points on the curve $C_\Gamma(\mathcal{Z}_\Gamma)$ that are symmetric with respect to the median, which in the dose space $\mathcal{Z}_\Gamma$ is given by $H^{-1}(0.5) = -\nu\zeta$. orangeConstructing this set in such a straightforward and elegant way is possible only if we consider the entire unrestricted dose space, meaning $\mathcal{Z}_\Gamma = \mathbb{R}$. While relying on an unbounded dose space may not directly translate into practically viable designs for actual experiments, it provides an essential theoretical benchmark against which restricted, more realistic designs can be evaluated.

Finally, the optimal design within this set is presented is characterized in the following theorem.

Theorem 3. *Consider an optimality criterion Φ that satisfies Condition 1 and assume a logistic or probit model in (1) under parameterization (2). Then, for a given value $\Gamma \in (0, 1)$, the optimal design has two support points $H^{-1}(0.5) \pm \delta$ where $\delta > 0$ and weights w and $1 - w$, with $w > 0$.*

Remark 2. *It is important to clarify that the geometric symmetry of the design is with respect to the median, not the target percentile. Specifically, the symmetry of the support points $H^{-1}(0.5) \pm \delta$ in the induced dose space $\mathcal{Z}_\Gamma$ implies that, when transformed back to the original experimental scale $\mathcal{X}$, the actual optimal doses remain equidistant from the median (i.e., the dose $\mu_{0.5}$ yielding a 50% probability of positive response). Consequently, whenever $\Gamma \neq 0.5$, the optimal design is inherently asymmetric with respect to the target dose μ_Γ.*

By Caratheodory's Theorem [see, for instance, Silvey (1980)], when working with a model with two parameters, the optimal design consists of at most three points. In the extreme situation, to compute the optimal design, we must optimize five parameters: three corresponding to the doses and two to their weights. However, under Theorem 3, the optimal design can be found by optimizing only two parameters because, since the doses are symmetric about the median, they depend only on the parameter δ (that represents the distance from the median) and the weight of each dose given by ω. Even though we determine the optimality criterion and we assume logistic or probit model in (1), an explicit analytical expression for the optimal design in Theorem 3 can not be given because of its dependency on Γ. However, given a value Γ, this dimensional reduction is useful to implement an efficient algorithm that provides explicit values for δ and w.

4 Computation of Optimal Designs

Using the structure provided by Theorem 3, we can simplify the optimization problem further analytically. Since the exact model depends on both Γ and W, we cannot provide a fixed numerical solution; however, we can reduce the complexity to a single dimension. In this section, for the A- and cc-optimality criteria, we express the optimal weight w as a function of δ (the detailed derivations are provided in Appendix B). Consequently, finding the optimal design reduces to maximizing a specific objective function with respect to the single parameter δ.

4.1 A-optimality

The A-optimality criterion minimizes the sum of the variance of both model parameter estimates. Mathematically, this corresponds to minimizing the trace of the inverse of the information matrix, $\mathrm{tr}[I(\xi)^{-1}]$. Following (5), the inverse of $\mathrm{I}(\xi)$ is

$$\mathrm{I}(\xi)^{-1} = \frac{1}{t_\Gamma v_\Gamma - u_\Gamma^2} \begin{pmatrix} v_\Gamma & -u_\Gamma \\ -u_\Gamma & t_\Gamma \end{pmatrix} (\xi).$$

Then,

$$\begin{aligned} \xi_A^* &= \mathrm{argmin}_{\xi \in \Xi_\Gamma} \left\{ Var(\widehat{\mu}_\Gamma \mid \xi) + Var(\widehat{\eta}_\Gamma \mid \xi) \right\} = \mathrm{argmin}_{\xi \in \Xi_\Gamma} \left\{ tr[I(\xi)^{-1}] \right\} \\ &= \mathrm{argmin}_{\xi \in \Xi_\Gamma} \left\{ \frac{v_\Gamma}{t_\Gamma v_\Gamma - u_\Gamma^2}(\xi) + \frac{t_\Gamma}{t_\Gamma v_\Gamma - u_\Gamma^2}(\xi) \right\} \\ &= \mathrm{argmax}_{\xi \in \Xi_\Gamma} \left\{ \frac{t_\Gamma v_\Gamma - u_\Gamma^2}{v_\Gamma + t_\Gamma}(\xi) \right\} = \mathrm{argmax}_{\xi \in \Xi_\Gamma} \left\{ \Phi_A(\xi) \right\}. \end{aligned} \tag{9}$$

Φ_A satisfies Condition 1 because the partial derivative of Φ_A with respect to v_Γ is strictly positive; so Theorem 3 applies and the A-optimal design is a two-point design in which the doses are symmetric around the median.

An aspect to consider when using this criterion is that if there is any difference in the magnitudes of the parameters' units of measure, the parameter with the larger units of measurement will dominate the optimization. In order to mitigate scale differences in parameter estimation, Dette (2002) works with a standardized information matrix (instead of the classical one) defined as

$$\widetilde{\mathrm{I}}(\xi) = diag(\sigma_1, \sigma_2)\mathrm{I}(\xi)diag(\sigma_1, \sigma_2),$$

where the values σ_1 and σ_2 are the the standard deviations of $\widehat{\mu}_\Gamma$ and $\widehat{\eta}_\Gamma$ obtained under their respective c-optimal designs. Flournoy et al. (2025) fully characterize these c-optimal designs for a wide catalog of models and obtain that $\sigma_1^2 = d_1^2/\eta_\Gamma^2$ and $\sigma_2^2 = \eta_\Gamma^2 d_2^2$ where d_1^2 and d_2^2 are constant values for the model chosen. Specifically, for the logistic model they are $(\pm 2.40 - \nu)\zeta$ and for the probit they are $(\pm 1.58 - \nu)\zeta$.

The design that optimizes the A-optimality criterion for this standardized information matrix is known as the standardized A-optimal design. Since the standardized matrix is a linear transformation of the original information matrix, the geometric procedure described in Section 3 applies, and the characterization of the standardized A-optimal design follows from Theorem 3.

Proposition 4. *In the conditions of Theorem 3, let $f_A(\delta) = d_1^2 + (\nu_\Gamma \zeta_\Gamma - \delta)^2 d_2^2$. Then the standardized A-optimal design is*

$$\xi_A^* = \begin{Bmatrix} H^{-1}(0.5) - \delta^* & H^{-1}(0.5) + \delta^* \\ \omega^* & 1 - \omega^* \end{Bmatrix}$$

where

$$\omega^* = \frac{\sqrt{f_A(\delta^*)}}{\sqrt{f_A(-\delta^*)} + \sqrt{f_A(\delta^*)}} \qquad \textit{and} \qquad \delta^* = \operatorname{argmax}_{\delta \in (0,\infty)} \frac{h(-\nu\zeta + \delta)\delta}{\sqrt{f_A(-\delta)} + \sqrt{f_A(\delta)}}.$$

In Table 2, standardized A-optimal designs are given for five different values of Γ. Due to the symmetry of the functions, the optimal design for $\Gamma > 0.5$ has the same doses as the optimal for $1 - \Gamma$ with the weights swapped.

Standard A-optimal		Logistic Toxicity	Logistic Weights	Probit Toxicity	Probit Weights
Γ	0.1	$(0.118, 0.882)$	$(0.682, 0.318)$	$(0.09, 0.91)$	$(0.688, 0.312)$
	0.2	$(0.146, 0.854)$	$(0.677, 0.323)$	$(0.111, 0.889)$	$(0.666, 0.334)$
	0.3	$(0.169, 0.831)$	$(0.635, 0.365)$	$(0.126, 0.874)$	$(0.62, 0.38)$
	0.4	$(0.184, 0.816)$	$(0.573, 0.427)$	$(0.136, 0.864)$	$(0.563, 0.437)$
	0.5	$(0.189, 0.811)$	$(0.5, 0.5)$	$(0.14, 0.86)$	$(0.5, 0.5)$

Table 2 Standardized A-optimal designs for the logistic and probit models with different Γ values.

4.2 cc-optimality

The cc-optimal design (Flournoy et al. 2025) prioritizes the estimation of μ_Γ over the estimation of η_Γ. To ensure the criterion is invariant to the scale of the parameters, it is defined in terms of efficiencies rather than raw variances. The efficiencies for each parameter are

$$\mathrm{Eff}_\mu(\xi) = \frac{\sigma_1^2}{Var(\widehat{\mu}_\Gamma \mid \xi)} \text{ and } \mathrm{Eff}_\eta(\xi) = \frac{\sigma_2^2}{Var(\widehat{\eta}_\Gamma \mid \xi)}. \tag{10}$$

where σ_1^2 and σ_2^2 are the minimum possible variances obtained with the respective c-optimal designs.

Formally, we define the unrestricted cc-optimal design ξ^*_{cc} as the solution to the following constrained optimization problem:

$$\xi^*_{cc} = \arg\max_{\xi \in \Xi_\Gamma} \mathrm{Eff}_\eta(\xi) \quad \text{subject to} \quad \mathrm{Eff}_\mu(\xi) \geq e_\mu.$$

where $e_\mu \in (0, 1)$ represents a user-defined minimum efficiency threshold for the target dose. It should be noted that in Flournoy et al. (2025), the cc-optimal design is additionally defined with a restriction on the dose space. In contrast, this work primarily focuses on the unrestricted case to fully exploit the geometric properties of the model, although restricted designs are also addressed in Section 4.3.

This approach allows the researcher to guarantee a specific level of performance for the target dose (e_μ) while optimizing the experimental effort to learn about the slope.

Following Flournoy et al. (2025), this constrained design can be solved by computing, for each $\lambda \in [0, 1]$,

$$\xi_{cc}^{\lambda} = \text{argmin}_{\xi \in \Xi_\Gamma} \left\{ \lambda \text{Eff}_\mu(\xi)^{-1} + (1 - \lambda)\text{Eff}_\eta(\xi)^{-1} \right\} \tag{11}$$

and selecting from $\{\xi_{cc}^{\lambda}\}_{\lambda \in [0,1]}$ the design that maximizes $\text{Eff}_\eta(\xi)$ while satisfying $\text{Eff}_\mu(\xi) > e_\mu$.

Observe that ξ_{cc}^{λ} can be written as

$$\begin{aligned} \xi_{cc}^{\lambda} = \text{argmin}_{\xi \in \Xi_\Gamma} \left(\lambda \frac{v_\Gamma}{(t_\Gamma v_\Gamma - u_\Gamma^2)\sigma_1^2} + (1 - \lambda) \frac{t_\Gamma}{(t_\Gamma v_\Gamma - u_\Gamma^2)\sigma_2^2} \right)(\xi) \\ = \text{argmax}_{\xi \in \Xi_\Gamma} \left(\frac{t_\Gamma v_\Gamma - u_\Gamma^2}{(1 - \lambda)\sigma_1^2 t_\Gamma + \lambda \sigma_2^2 v_\Gamma} \right)(\xi), \end{aligned}$$

where the last objective function satisfies Condition 1. So Theorem 3 applies to ξ_{cc}^{λ} and consequently to ξ_{cc} too.

Proposition 5. *Let $f_{cc}^{\lambda}(\delta) = (1 - \lambda)d_1^2 + (\nu\zeta - \delta)^2 \lambda d_2^2$, the Φ_{cc}^{λ}-optimal design is*

$$\xi^* = \begin{Bmatrix} -\nu\zeta - \delta^* & -\nu\zeta + \delta^* \\ \omega^* & 1 - \omega^* \end{Bmatrix},$$

where

$$\omega^* = \frac{\sqrt{f_{cc}^{\lambda}(\delta^*)}}{\sqrt{f_{cc}^{\lambda}(-\delta^*)} + \sqrt{f_{cc}^{\lambda}(\delta^*)}} \quad \text{and} \quad \delta^* = \underset{\delta \in (0,\infty)}{\text{argmax}} \frac{h(-\nu\zeta + \delta)\delta}{\sqrt{f_{cc}^{\lambda}(-\delta)} + \sqrt{f_{cc}^{\lambda}(\delta)}}.$$

In Table 3 the cc-optimal design with different values of Γ and e_μ values are given. As with the A-optimal criteria, the optimal design for $\Gamma > 0.5$ can be computed by swapping the weights of the optimal design for $1 - \Gamma$.

cc-optimal		Logistic		Probit	
$e_\mu = 0.80$		Toxicity	Weights	Toxicity	Weights
Γ	0.1	$(0.125, 0.875)$	$(0.805, 0.195)$	$(0.104, 0.896)$	$(0.847, 0.153)$
	0.2	$(0.183, 0.817)$	$(0.851, 0.149)$	$(0.16, 0.84)$	$(0.834, 0.166)$
	0.3	$(0.243, 0.757)$	$(0.804, 0.196)$	$(0.208, 0.792)$	$(0.764, 0.236)$
	0.4	$(0.289, 0.711)$	$(0.688, 0.312)$	$(0.241, 0.759)$	$(0.65, 0.35)$
	0.5	$(0.307, 0.693)$	$(0.5, 0.5)$	$(0.253, 0.747)$	$(0.5, 0.5)$

Table 3 cc-optimal designs with $e_\mu = 0.85$ for the logistic and probit models with different Γ values.

4.3 Comments on Restricted Dose Spaces

Propositions 4 and 5 present an efficient method for deriving A- and cc-optimal designs assuming an unrestricted dose space. This efficiency is particularly valuable because the framework addresses an arbitrary $\Gamma \in (0, 1)$; a researcher may require a target percentile not explicitly listed in Tables 2 and 3. In such cases, the ability to rapidly compute the optimal design is essential.

This requirement becomes even more critical for the cc-optimal design. Since it is a compound design, finding the solution requires computing optimal designs across a range of $\lambda \in (0, 1)$ values to select the best. Consequently, determining the cc-optimal design involves solving a series of optimization problems rather than a single one. Given this iterative process, any reduction in computational complexity significantly enhances the overall efficiency of the algorithm.

Nevertheless, these unrestricted results have a clear limitation: as they are symmetric above the median, they always involve a very high dose [i.e., a dose for which $H(x) > 0.5$], making them impractical in many contexts. This requires restricting the dose space. Furthermore, Flournoy et al. (2025) define the cc-optimal design in a restricted dose space. Therefore, to provide a complete solution for this criterion, the restricted design problem must be addressed.

Biedermann et al. (2006) characterize the restricted optimal designs for a class of criteria defined by Kiefer (1974), which includes the cc-optimal design, and different restrictions over the dose space, In order to apply the results by Biedermann et al. (2006), the function h_Γ must satisfy some conditions that are ensured for the logistic and probit models. Following their work, the optimal design under the restriction $(-\infty, \Delta]$ is characterized as:

i) The unrestricted optimal design provided its support points lie entirely within $(-\infty, \Delta]$.
ii) A two-point design where the upper dose is fixed at the boundary Δ.

The first case corresponds to the optimization problem already solved in propositions 4 and 5. A similar univariate optimization approach can be employed for the second case for both A- and cc-optimal designs (derivations detailed are provided in Appendix C).

Proposition 6. *Let $g_A(z) = h^2(z)(d_1^2 + z^2 d_2^2)$. The A-optimal design under the restricted space $(-\infty, \Delta]$ and case (ii) is*

$$\xi_m^* = \begin{Bmatrix} z^* & \Delta \\ w^* & 1 - w^* \end{Bmatrix},$$

where

$$w^* = \frac{\sqrt{g_A(\Delta)}}{\sqrt{g_A(z)} + \sqrt{g_A(\Delta)}} \text{ and } z^* = \underset{z \in (-\infty, \Delta)}{\text{argmax}} \frac{h_\Gamma(z)(\Delta - z)}{\sqrt{g_A(\Delta)} + \sqrt{g_A(z)}}.$$

Proposition 7. *Let* $g_{cc}^{\lambda}(z) = h_{\Gamma}^{2}(z)\big[(1-\lambda)d_1^2 + \lambda z^2 d_2^2\big]$. *The cc-optimal design under the restricted space* $(-\infty, \Delta]$ *and case (ii) is*

$$\xi_m^* = \begin{Bmatrix} z^* & \Delta \\ w^* & 1-w^* \end{Bmatrix},$$

where

$$w^* = \frac{\sqrt{g_{cc}^{\lambda}(\Delta)}}{\sqrt{g_{cc}^{\lambda}(z^*)} + \sqrt{g_{cc}^{\lambda}(\Delta)}} \text{ and } z^* = \underset{z\in(-\infty,\Delta)}{\operatorname{argmax}} \frac{h_{\Gamma}(z)(\Delta - z)}{\sqrt{g_{cc}^{\lambda}(\Delta)} + \sqrt{g_{cc}^{\lambda}(z)}}.$$

5 A Comparative Simulation Study

The main goal of this section is to ascertain the practical impact of the designs introduced in Section 4.1. To this end, a comparative study is carried out between the models introduced in Section 4.1 and a selected group of referential designs, see Table 4. First, the theoretical efficiencies of each design with respect to the c-optimal design are graphically compared. Then, the finite-sample performance of MLEs from the new and the classical designs are compared all assuming correct model specification. A logistic model, $f(x) = [1 + exp(-x)]^{-1}$, is considered throughout the study.

Abbreviation	Description
Aopt	Standardized A-optimal design for $\Gamma \neq 0.5$ in Proposition 4, Table 2
Acls*	Standardized A-optimal design, Dette and Sahm (1997).
ccopt	cc-optimal design by Flournoy et al. (2025); Proposition 5, Table 3 with $\mathrm{Eff}_{\mu}(\xi) = 0.8$.
dopt	The D-optimal design; p. 155 in Fedorov and Leonov (2014).
unif	The uniform design allocating equal proportions of experimental units to 10 equally spaced doses in the interval $[F^{-1}(0.1), F^{-1}(0.9)]$.

*Acls is Aopt for $\Gamma = 0.5$ in Proposition 4

Table 4 Designs used in the comparative study.

Figure 5 displays the theoretical efficiencies (10) as functions of Γ. By construction, the ccopt requires efficiency of the MLE at the target dose to be at least 0.8 regardless the value of Γ as shown in Figure 5, left panel. This leads to relatively low efficiencies for the slope MLE, but these efficiencies increase as the restriction is relaxed below 0.8.

Aside from the ccopt which was just discussed, the Aopt achieves the highest relative efficiencies for both target and slope MLEs as Γ approaches more extreme values. As expected, the Aopt approaches Acls design as Γ values are close to 0.5, where it and the other designs all exhibit very similar efficiencies.

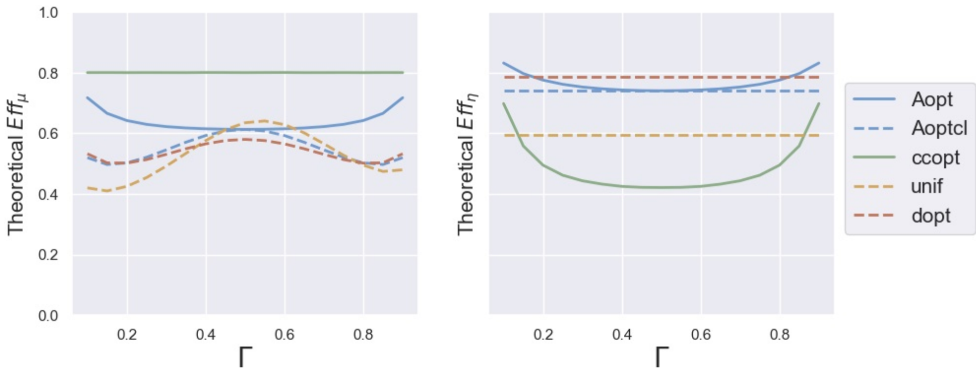


Fig. 5 Theoretical efficiencies of MLEs of the target dose (left panel) and slope (right panel) for various designs relative to the c-optimal design for the target dose (left) and slope (right).

The choice of an optimality criterion strongly depends on the objective of the study. c-optimal designs for the target dose are generally not useful for estimation purposes because they are often singular [in the logistic model, this occurs when $\Gamma \in (0.083, , 0.917)$; see Ford et al. (1992)] and we have to estimate two values, intercept and slope, of the linear relationship with the dose. Nevertheless, they serve as important benchmarks for evaluating efficiencies. Standardized A-optimal designs are useful for the joint estimation of both parameters, assigning them comparable precision because they are on a similar scale. cc-optimal designs minimize the variance of the slope MLE while guaranteeing a minimum efficiency for the target-dose MLE. D-optimal designs minimize the volume of the confidence ellipsoid for target and slope MLEs; useful for the joint estimation but equal precision is not controlled.

For the sake of brevity, the simulation study reported in Table 5 considers neither an adaptive framework to initialize the experiment nor a restricted dose space to address the ethical or economic concerns associated with the use of extreme doses. Instead, the study focuses on evaluating the performance of the designs in Table 4 for small to moderate sample sizes, $n = 25$, 50, 100.

The existence of the MLE requires that both successes and failures overlap within the set of administered doses [see Silvapulle (1981)]. In our context, when the slope around the target dose is steep, separability issues may arise for small to moderate sample sizes. Nevertheless, Flournoy et al. (2020) reports that, for slopes classified as moderate, the MLE exists with probability at least 90% for sample sizes smaller than 20. Therefore, we fix $\Gamma = 0.3$, which yields $\mu_{0.3} = -0.847$ and $\eta_{0.3} = 0.21$. According to Flournoy et al. (2020), this corresponds to a moderate slope.

For each design, ξ, the simulation procedure iterates 10000 times the allocation of n individuals. In each iteration, $i = 1, \cdots, 10000$, a random sequence of n binary responses is randomly generated with the logistic model. Then, the MLEs $\hat{\mu}_{n,i}$ and $\hat{\eta}_{n,i}$ are estimated. The sample variances of these estimator sets are denoted, respectively, $\widehat{var}(\hat{\mu}_n)$ and $\widehat{var}(\hat{\eta}_n)$. For each design, ξ, of Table 4 and sample size, n, we report in

Table 5 the estimated efficiencies of (10) that are obtained as follows

$$\widehat{\text{Eff}}_{n,\mu}(\xi) = \frac{\sigma_1^2}{\widehat{var}(\widehat{\mu}_n)} \text{ and } \widehat{\text{Eff}}_{n,\eta}(\xi) = \frac{\sigma_2^2}{\widehat{var}(\widehat{\eta}_n)}. \tag{12}$$

Observe that, for all the designs compared, as n grows the efficiencies approach the theoretical values. The slope MLE shows large variability (with respecto to the variability of the c-optimal) for $n = 25, 50$ but this relative instability gets dramatically reduced when $n = 100$. However, target-dose MLE is more stable even for $n = 25$ and the estimated efficiency approaches 0.5 value for $n = 50$. Observe that the Aopt design, aside the ccopt, reaches the largest target-dose estimated efficiencies whereas for the slope is not quite competitive for small values of n.

	n=25		n=50		n=100		Theoretical	
	$\mu_{0.3}$	$\eta_{0.3}$	$\mu_{0.3}$	$\eta_{0.3}$	$\mu_{0.3}$	$\eta_{0.3}$	$\mu_{0.3}$	$\eta_{0.3}$
Aopt	0.375	0.015	0.513	0.039	0.586	0.439	0.622	0.753
Aoptcl	0.307	0.016	0.430	0.095	0.501	0.570	0.545	0.74
ccopt	0.564	0.010	0.626	0.019	0.708	0.159	0.8	0.443
unif	0.365	0.060	0.424	0.379	0.468	0.503	0.492	0.594
dopt	0.265	0.016	0.407	0.072	0.490	0.606	0.531	0.787

Table 5 Mean a-posteriori efficiency, with respect to the c-optimal for each n.

6 Discussion

This work widens the body of knowledge concerning optimal designs for the frequently used binary response studies that aim to estimate a dose μ_Γ with a prespecified positive response rate Γ.

A relevant work in this context is Ford et al. (1992) who, using geometrical procedures, fully characterized c-optimal and D-optimal designs focusing on estimation of μ_Γ alone. The importance of η_Γ, the slope of the response function at the target dose, was recognized long ago in Wetherill (1963); but we feel it has not received the attention it deserves in the literature. As a matter of fact, typically, estimates of μ_Γ are found by an approximation method that combines the estimates of the parameters $\boldsymbol{\theta}$ chosen in model (1); see Wu (1988).

In the same vein as Flournoy et al. (2025), we study the joint estimation of both parameters. We introduce a parameterization in Section 2 for $\boldsymbol{\theta} = (\mu_\Gamma, \eta_\Gamma)$ and any $\Gamma \in (0, 1)$ whose main benefit is that the information matrix depends directly on the parameters of interest and the optimization in the new dose space does not depend on the parameters. However, symmetrical properties that appear in the geometrical reasoning in Ford et al. (1992) no longer hold. Biedermann et al. (2006), Sitter and Fainaru (1997) and Dette and Sahm (1998) use the same parameterization as in Section 2, but they only consider $\Gamma = 0.5$. This choice allows them to maintain symmetrical properties in their geometrical reasoning to obtain optimal designs; but then estimation of μ_Γ and η_Γ is made as a transformation of $\mu_{0.5}$ and $\eta_{0.5}$ estimators.

In Section 3, the geometrical procedure in Sitter and Fainaru (1997) (focused on the estimation of $\mu_{0.5}$ and $\eta_{0.5}$) is generalized to estimate μ_Γ and η_Γ for any value $\Gamma \in (0, 1)$. In Theorem 3, we show that the optimal design is a two-point design in which the doses are symmetric about the median regardless of the Γ value of interest. This discovery simplifies the optimization task, reducing the computational complexity from a five-parameter optimization problem to one of a single-parameter.

In Section 4, explicit mathematical expressions are given for the standardized A- and cc-optimal designs introduced in Flournoy et al. (2025) by applying Theorem 3. Section 4.3 connects this work with Biedermann et al. (2006) where the dose space is restricted. Restricted dose spaces overcome the usual requirement of extreme doses in optimal designs. For example, in toxicological studies, ethical concerns arise when individuals are exposed to extreme doses, as these may be harmful to patients.

Finally, Section 5 presents a comparative study of the new designs introduced in Section 4. The limited scope of the simulation study highlights several limitations of this work that warrant further research. First, the robustness to misspecification and parameter uncertainty is an important issue not considered in this work. However, efficiency study for the probit model shows similar results (not included in the report). Parameter uncertainty in binary designs has been widely treated in the literature and most of the research advice for the use of adaptive or sequential design approaches to overcome the problem [see Fedorov and Leonov (2014), Dette et al. (2013)] and algorithms to this end have been proposed in the literature, Yang et al. (2013). Second, ML estimation can have non-existence problems or bias specially for small sample sizes, see Flournoy et al. (2020) and the references therein. Estimation packages incorporate corrections that mitigate those problems, see Firth (1993) or Kosmidis (2014), but a specific study of these situations in our context would widen the applicability of the proposed designs.

It is important to clarify that the decision to address the estimation problem in 1 by revisiting and extending the geometric framework in Sitter and Fainaru (1997) is not trivial. Although these designs could have been obtained using purely analytical or computational approaches yielding equally valid results, such methods often function as "black boxes" that obscure the underlying structure of the solution. In contrast, the geometric perspective presented here provides a mechanistic understanding of the problem. By visualizing the information space, we can explicitly observe how designs are constructed through convex combinations of elementary designs, revealing the specific contribution of each dose to the total information and providing a comprehensive view of the matrices of all possible designs. This offers a distinct advantage over other geometric methods, such as those by Elfving (1952); Ford et al. (1992); Biedermann et al. (2006), which are typically restricted to two-dimensional representations. Furthermore, this framework establishes a basis for addressing complex problems that were omitted from the main text due to space constraints, such as modeling information accumulation in sequential designs.

It is worth emphasizing that most of the designs presented in Table 4 can not be implemented directly in practice. As noted in Ford et al. (1992), such designs are valuable as theoretical benchmarks for practical designs. Examples of estimating and applying optimal designs sequentially are described in Chapter 8 in Fedorov and

Leonov (2014); Lane et al. (2014) describes how much can be accomplished with just two-stages. But these do not directly apply to the dose-response models that are central to this paper. An up-and-down dose-finding design can center doses around any unknown percentile and hence can be used directly to obtain observations near the unknown optimal ones or to serve as a start-up for a model-based procedures. A good choice might be the biased-coin up-and-down design that rapidly centers allocations around any unknown percentile; see Oron and Flournoy (2024).

Another approach is to begin with a start-up design [see Section 3.5 in Flournoy et al. (2020)] that provides a fixed amount of information, represented geometrically as a point $M_0 = (t_0, u_0, v_0)$ in the information space. The challenge for the subsequent stage is not to find a globally optimal design from scratch [see O'Brien (1995), ch. 19 Atkinson et al. (2007)], but to find a design M_{new} such that the convex combination of both stages, $M_{total} = kM_{new} + (1-k)M_0$, maximizes the optimality criterion.

Although analytical approaches often treat this as a complex constrained optimization problem involving derivatives of a modified criterion, our geometric framework allows for a natural interpretation: the accumulated information effectively "shifts" the center of the optimization contours in the design space. Consequently, optimizing the next stage reduces to maximizing the shifted criterion:

$$\phi^*(t, u, v) = \phi\big[kt + (1-k)t_0,\ ku + (1-k)u_0,\ kv + (1-k)v_0\big].$$

ϕ^* satisfies Condition 1 provided that ϕ does. Therefore, Theorem 3 remains applicable, significantly simplifying the optimization problem. We plan to exploit this computational advantage in future work to characterize optimal multi-stage designs.

Code Availability

The Python implementation of the algorithms aimed at computing the standardized A-optimal and cc-optimal designs described in this paper is available at https://doi.org/10.5281/zenodo.18611357. This repository includes tutorials and the scripts necessary to reproduce the results presented in Tables 2 and 3.

References

Atkinson, A., Donev, A., Tobias, R.: Optimum Experimental Designs, With SAS. OUP Oxford, Oxford (2007)

Biedermann, S., Dette, H., Zhu, W.: Optimal designs for dose–response models with restricted design spaces. Journal of the American Statistical Association **101**(474), 747–759 (2006)

Christensen, D., Novik, G.P., Unneberg, E.: Estimating sensitivity with the bruceton method: Setting the record straight. Propellants, Explosives, Pyrotechnics **49**(7), 202400022 (2024) https://doi.org/10.1002/prep.202400022

Chang, H., Ying, Z.: Nonlinear sequential designs for logistic item response theory models with applications to computerized adaptive tests. The Annals of Statistics **37**(3), 1466–1488 (2009) https://doi.org/10.1214/08-AOS614

Dette, H., Bornkamp, B., Bretz, F.: On the efficiency of two-stage response-adaptive designs. Statistics in Medicine **32**(10), 1646–1660 (2013) https://doi.org/10.1002/sim.5555

Dette, H.: Designing experiments with respect to 'standardized' optimality criteria. Journal of the Royal Statistical Society: Series B (Methodological) **59**(1), 97–110 (2002) https://doi.org/10.1111/1467-9868.00056

Dette, H., Sahm, M.: Standardized optimal designs for binary response experiments. South African Statistical Journal **31**(2), 271–298 (1997)

Dette, H., Sahm, M.: Minimax optimal designs in nonlinear regression models. Statistica Sinica, 1249–1264 (1998)

Elfving, G.: Optimum allocation in linear regression theory. The Annals of Mathematical Statistics, 255–262 (1952)

Fedorov, V.: Theory of Optimal Experimental Designs. Academic Press, New York (1972)

Fedorov, V., Gaivoronski, A.: Design of experiments under constraints. Working Paper WP-84-8, International Institute for Applied Systems Analysis (IIASA), Laxenburg, Austria (1984)

Firth, D.: Bias reduction of maximum likelihood estimates. Biometrika **80**(1), 27–38 (1993)

Fedorov, V., Leonov, S.L.: Optimal Design for Nonlinear Response Models. CRC Press, Boca Raton (2014)

Flournoy, N., Moler, J., Plo, F.: Performance measures in dose-finding experiments. International Statisticial Review **88**, 728–751 (2020) https://doi.org/10.1111/insr.12363

Flournoy, N., Moler, J., Sada, M.: Influence of the slope in percentile estimation through binary regression for dose-finding experiments. Statistics **59**(2), 305–329 (2025)

Ford, I., Torsney, B., Wu, C.J.: The use of a canonical form in the construction of locally optimal designs for non-linear problems. Journal of the Royal Statistical Society Series B: Statistical Methodology **54**(2), 569–583 (1992)

Kiefer, J.: General equivalence theory for optimum designs (approximate theory). The Annals of Statistics, 849–879 (1974)

Kosmidis, I.: Bias in parametric estimation: reduction and useful side-effects. WIREs Computational Statistics **6**(3), 185–196 (2014) https://doi.org/10.1002/wics.1296

Leek, M.R.: Adaptive procedures in psychophysical research. Perception & Psychophysics **63**, 1279–1292 (2001) https://doi.org/10.3758/BF03194543

Lagoda, T., Sonsino, C.: Comparison of different methods for presenting variable amplitude loading fatigue results. Material Science & Engineering Technology **35**(1), 13–20 (2004) https://doi.org/10.1002/mawe.200300692

Lane, A., Yao, P., Flournoy, N.: Information in a two-stage adaptive optimal design. Journal of Statistical Planning and Inference **144**, 173–187 (2014)

Oron, A.P., Flournoy, N.: Up-and-down: The most popular, most reliable, and most overlooked dose-finding design. The New England Journal of Statistics in Data Science **3**(3), 222–233 (2024)

Oron, A.P., Souter, M., Flournoy, N.: Understanding research methods: Up-and-down designs for dose-finding. Anesthesiology **137**(2), 137–150 (2022) https://doi.org/10.1097/ALN.0000000000004282

O'Brien, T.E.: Optimal design and lack of fit in nonlinear regression models. Statistical Modelling, (eds Seeber, G., Francis, B., Hatzinger, R. Steckel-Berger, G.), 201–206 (1995) https://doi.org/10.1007/978-1-4612-0789-4_25

Pollak, R., Palazotto, A., Nicholas, T.: A simulation-based investigation of the staircase method for fatigue strength testing. Mechanics of Materials **38**(12), 1170–1181 (2006) https://doi.org/10.1016/j.mechmat.2005.12.005

Sitter, R.R., Fainaru, I.: Optimal designs for the logit and probit models for binary data. Canadian Journal of Statistics **25**(2), 175–190 (1997)

Silvey, S.D.: Optimal Design: An Introduction to the Theory for Parameter Estimation. Monographs on Applied Probability and Statistics, vol. 14. Chapman and Hall, London (1980)

Silvapulle, M.J.: On the existence of maximum likelihood estimators for the binomial response models. Journal of the Royal Statistical Society. Series B (Methodological). **43**(3), 310–313 (1981)

Sitter, R., Wu, C.: Optimal designs for binary response experiments: Fieller, D, and A criteria. Scandinavian Journal of Statistics, 329–341 (1993)

Wetherill, G.: Sequential estimation of quantal response curves. Journal of the Royal Statistical Society Series B: Statistical Methodology **25**(1), 1–38 (1963)

Wetherill, G.B., Levitt, H.: Sequential estimation of points on a psychometric function. British Journal of Mathematical and Statistical Psichology **18**(1), 1–10 (1965) https:

//doi.org/10.1111/j.2044-8317.1965.tb00689.x

Wu, D.F.J.: Optimal design for percentile estimation of a quantal response curve. Optimal Design and Analysis of Experiments (eds Dodge, Y., Fedorov, V.V. and Wynn, H.P.), 213–223 (1988)

Yang, M., Biedermann, S., Tang, E.: On optimal designs for nonlinear models: A general and efficient algorithm. Journal of the American Statistical Association **108**(504), 1411–1420 (2013) https://doi.org/10.1080/01621459.2013.806268

Appendix A Construction of the Right Boundary of the Convex Hull

To construct the right boundary of the convex hull, start from the set $\mathcal{CH}$ shown in Figure A1(a). Recall the axes in this figure are the elements of the information matrix given in Section 3.1: $t_\Gamma(\xi) = \eta_\Gamma^2 \sum_{i=1}^k \omega_i \; h^2(z_i)$; $u_\Gamma(\xi) = -\sum_{i=1}^k \omega_i \; h^2(z_i) \; z_i$; and $v_\Gamma(\xi) = \dfrac{1}{\eta_\Gamma^2} \sum_{i=1}^k$ where $h^2(z) = w^2(s)/\zeta_\Gamma^2$, for $w^2(s)$ defined in Table 1. Suppose we fix specific values $t_1 \in t_\Gamma(\Xi_\Gamma)$ and $u_1 \in u_\Gamma(\Xi_\Gamma)$, and consider the subset of points (t, u, v) contained in $\mathcal{CH}$.

In Figure A1 this set would be the dashed line drawn in Figure A1(b). The optimal design in this restricted set would be

$$\underset{(t_1,u_1,v)\in\mathcal{CH}_\Gamma}{\operatorname{argmax}} \; \{\phi(t_1, u_1, v)\} \tag{A1}$$

which is equivalent to $\underset{(t_1,u_1,v)\in\mathcal{CH}_\Gamma}{\operatorname{argmax}} \; \{v\}$. when criterion ϕ satisfies Condition 1. Similarly, if we only fix the value u_1, the optimal design under restriction $u_\Gamma(\xi^*) = u_1$ is $\bigcup_{t\in t_\Gamma(\Xi_\Gamma)} \underset{(t,u_1,v)\in\mathcal{CH}_\Gamma}{\operatorname{argmax}} \; \{v\}$, which is shown in Figure A1(c) as a gray curve. Finally, the optimal design without restriction is in the Right Boundary shown in Figure A1(d): $\bigcup_{\substack{t\in t_\Gamma(\Xi_\Gamma)\\ u\in u_\Gamma(\Xi_\Gamma)}} \underset{(t,u,v)\in\mathcal{CH}_\Gamma}{\operatorname{argmax}} \; \{v\}$.

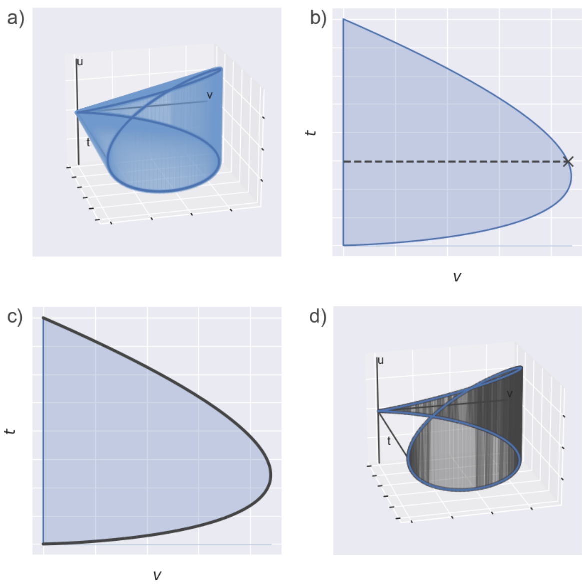


Fig. A1 The construction process of $\mathcal{RB}_\Gamma$ is illustrated for the logistic function with $\Gamma = 0.5$.

Appendix B Computation of Unrestricted Optimal Designs

In order to optimize the A- and cc-optimality criteria, we express the information matrix for designs characterized by Theorem 3. To simplify the notation, given the large number of formulae involved, we drop the subscript Γ from the notation.

B.1 Information Matrix under Theorem 3

Theorem 3 is useful because it makes the calculation of the optimal design easier, as the design and consequently the information matrix will depend only on two parameters. In order to compute the optimal design, it is necessary to give an expression of the information matrix and its standardized version.

Proposition 8. *Given the design*

$$\xi_{\delta,\omega} = \begin{Bmatrix} -\nu\zeta - \delta & -\nu\zeta + \delta \\ \omega & 1-\omega \end{Bmatrix} \tag{B2}$$

in the dose space $\mathcal{Z}$, the classical and standardized information matrices are, respectively,

$$\mathrm{I}(\xi_{\delta,\omega}) = \begin{pmatrix} \eta^2 h_\delta^2 & u_{\delta,\omega} \\ u_{\delta,\omega} & \dfrac{v_{\delta,\omega}}{\eta^2} \end{pmatrix} \quad \text{and} \quad \widetilde{\mathrm{I}}(\xi_{\delta,\omega}) = \begin{pmatrix} d_1^2\, h_\delta^2 & d_1 d_2\, u_{\delta,\omega} \\ d_1 d_2\, u_{\delta,\omega} & d_2^2\, v_{\delta,\omega} \end{pmatrix},$$

where $h_\delta^2 = h^2(-\nu\zeta - \delta)$, $u_{\delta,\omega} = \big(2\delta\omega + \nu\zeta - \delta\big)h_\delta^2$, $v_{\delta,\omega} = \big[\omega 4\nu\zeta\delta + (-\nu\zeta+\delta)^2\big]\dfrac{h_\delta^2}{\eta^2}$ and d_1^2 and d_2^2 are the values given in Flournoy et al. (2025).

Proof To begin, observe that

$$h^2(-\nu\zeta+\delta) = \frac{1}{\zeta^2}w^2\left(\frac{\delta}{\zeta}\right) = \frac{1}{\zeta^2}w^2\left(-\frac{\delta}{\zeta}\right) = h^2(-\nu\zeta-\delta).$$

To ease the notation, define $h_\delta^2 = h^2(-\nu\zeta \pm \delta)$.

From (5,6),

$$\begin{aligned}
t(\xi_{\delta,\omega}) &= \eta^2\big[\omega h_\delta^2 + (1-\omega)h_\delta^2\big] && = \eta^2 h_\delta^2 \\
u(\xi_{\delta,\omega}) &= -\big[(-\nu\zeta-\delta)\omega h_\delta^2 + (-\nu\zeta+\delta)(1-\omega)h_\delta^2\big] && = \big(2\delta\omega + \nu\zeta - \delta\big)h_\delta^2 \\
v(\xi_{\delta,\omega}) &= \frac{1}{\eta^2}\big[(-\nu-\delta)^2\omega h_\delta^2 + (-\nu\zeta+\delta)^2(1-\omega)h_\delta^2\big] && = \big[4\omega\nu\zeta\delta + (-\nu\zeta+\delta)^2\big]\frac{h_\delta^2}{\eta^2}
\end{aligned}$$

Then, defining $u_{\delta,w} = \big(2\delta\omega + \nu\zeta - \delta\big)h_\delta^2$ and $v_{\delta,\omega} = \big[4\omega\nu\zeta\delta + (-\nu\zeta+\delta)^2\big]h_\delta^2$, the information matrix is $\mathrm{I}(\xi_{\delta,\omega}) = \begin{pmatrix} \eta^2 h_\delta^2 & u_{\delta,\omega} \\ u_{\delta,\omega} & \dfrac{v_{\delta,\omega}}{\eta^2} \end{pmatrix}$.

Using the information matrix $\mathrm{I}(\xi_{\delta,\omega})$ and following the standardization process presented in Dette and Sahm (1997) and described in Section 4.1, the standardized information matrix is

$$\widetilde{\mathrm{I}}(\xi_{\delta,\omega}) = \begin{pmatrix} \frac{d_1}{\eta} & 0 \\ 0 & d_2\eta \end{pmatrix} \mathrm{I}(\xi_{\delta,\omega}) \begin{pmatrix} \frac{d_1}{\eta} & 0 \\ 0 & d_2\eta \end{pmatrix} = \begin{pmatrix} d_1^2\, h_\delta^2 & d_1 d_2\, u_{\delta,\omega} \\ d_1 d_2\, u_{\delta,\omega} & d_2^2\, v_{\delta,\omega} \end{pmatrix}.$$

□

B.2 A-Optimal Design

The standardized A-optimal design will be the one that maximizes the A-optimal criterion defined in (9) under the standardized information matrix:

$$\xi_A^* = \underset{\delta,\omega}{\operatorname{argmax}} \frac{(tv-u^2)d_1^2 d_2^2}{d_1^2 t + d_2^2 v}(\xi_{\delta,\omega}) = \underset{\delta,\omega}{\operatorname{argmax}} \frac{tv-u^2}{d_1^2 t + d_2^2 v}(\xi_{\delta,\omega}).$$

Observe how

$$\begin{aligned} (tv-u^2)(\xi_{\delta,\omega}) &= \left[4\omega\delta\nu\zeta + (\nu\zeta-\delta)^2 - (2\omega\delta+\nu\zeta-\delta)^2\right] h_\delta^4 \\ &= \left[4\omega\delta\nu\zeta + (\nu\zeta-\delta)^2 - 4\omega^2\delta^2 - 4\omega\delta\nu\zeta + 4\omega\delta^2 - (\nu\zeta-\delta)^2\right] h_\delta^4 \\ &= 4(1-\omega)\omega h_\delta^4 \delta^2. \end{aligned}$$

Then,

$$\begin{aligned} \xi_A^* &= \underset{\delta,\omega}{\operatorname{argmax}} \frac{4(1-\omega)\omega h_\delta^2\delta^2}{d_1^2 + [4\delta\nu\zeta\omega + (\delta-\nu\zeta)^2]d_2^2} \\ &= \underset{\delta,\omega}{\operatorname{argmax}} \frac{4(1-\omega)\omega h_\delta^2\delta^2}{[f_A(-\delta)-f_A(\delta)]\omega + f_A(\delta)} = \underset{\delta,\omega}{\operatorname{argmax}}\, \varphi_A(\delta,\omega), \end{aligned}$$

where $f_A(\delta) = d_1^2 + (\delta-\nu\zeta)^2 d_2^2$. To maximize $\varphi_A(\delta,\omega)$, we first compute the value of ω that maximizes it. To do this, we differentiate $\varphi_A(\delta,\omega)$ with respect to ω and set it equal to 0:

$$\frac{\partial\varphi_A}{\partial\omega} = 4h_\delta^2\delta^2 \frac{[-f_A(-\delta)+f_A(\delta)]\omega^2 - 2f_A(\delta)\omega + f_A(\delta)}{\Big([f_A(-\delta)-f_A(\delta)]\omega + f_A(\delta)\Big)^2} = 0.$$

This is only possible if $[-f_A(-\delta)+f_A(\delta)]\omega^2 - 2f_A(\delta)\omega + f_A(\delta) = 0$ which implies that

$$\omega = \frac{-f_A(\delta) \pm \sqrt{f_A(\delta)f_A(-\delta)}}{f_A(-\delta)-f_A(\delta)}.$$

Finally, φ_A will be maximal at the positive solution so

$$\omega^* = \frac{\sqrt{f_A(\delta) f_A(-\delta)} - f_A(\delta)}{f_A(-\delta) - f_A(\delta)} = \frac{\sqrt{f_A(\delta)}}{\sqrt{f_A(-\delta)} + \sqrt{f_A(\delta)}} \in (0,1).$$

Then,

$$\begin{aligned}
\xi_A^* &= \operatorname*{argmax}_{\delta} 4h_\delta^2\delta^2\ (1-\omega)\omega\left(\left[f_A(\delta) - f_A(-\delta)\right]\omega + f_A(\delta)\right)^{-1} &\text{(B3)}\\
&= \operatorname*{argmax}_{\delta} 4h_\delta^2\delta^2\ \frac{\sqrt{f_A(\delta)f_A(-\delta)}}{\left[\sqrt{f_A(-\delta)} + \sqrt{f_A(\delta)}\right]^2}\left[\sqrt{f_A(-\delta)f_A(\delta)} - f_A(\delta) + f_A(\delta)\right]^{-1} &\\
&= \operatorname*{argmax}_{\delta} \frac{4h_\delta^2\delta^2}{\left[\sqrt{f_A(-\delta)} + \sqrt{f_A(\delta))}\right]^2} = \operatorname*{argmax}_{\delta} \frac{h_\delta\delta}{\sqrt{f_A(-\delta)} + \sqrt{f_A(\delta)}}. &\text{(B4)}
\end{aligned}$$

Therefore, the optimization problem is reduced to optimizing (B4): $\operatorname{argmax}_\delta h_\delta\delta/[\sqrt{f_A(-\delta)} + \sqrt{f_A(\delta)}]$, which can be calculated computationally in a short time.

B.3 cc-Optimality

The cc-optimality criteria must be solved for each $\lambda \in (0,1)$. The cc-optimal design given λ is

$$\begin{aligned}
\xi_{cc}^\lambda &= \operatorname*{argmax}_{\delta,\omega} \frac{tv - u^2}{(1-\lambda)\sigma_1^2 t + \lambda\sigma_2^2 v}(\xi_{\delta,\omega}) = \operatorname*{argmax}_{\delta,\omega} \frac{t_{\delta,\omega}v_{\delta,\omega} - u_{\delta,\omega}^2}{(1-\lambda)d_1^2 t_{\delta,\omega} + \lambda d_2^2 v_{\delta,\omega}}\\
&= \operatorname*{argmax}_{\delta,\omega} \frac{4\omega(1-\omega)h_\delta^2\delta^2}{\left[f_{cc}^\lambda(-\delta) - f_{cc}^\lambda(\delta)\right]\omega + f_{cc}^\lambda(\delta)},
\end{aligned}$$

where $f_{cc}^\lambda(\delta) = (1-\lambda)d_1^2 + (\nu\zeta - \delta)^2\lambda d_2^2$. Observe that the function to maximize and $\varphi_A(\delta,\omega)$ are analogous. For this reason, the analytical calculus are similar and the optimal values for ω and δ are

$$\omega^* = \frac{\sqrt{f_{cc}^\lambda(\delta^*)}}{\sqrt{f_{cc}^\lambda(-\delta^*)} + \sqrt{f_{cc}^\lambda(\delta^*)}} \quad \text{and} \quad \delta^* = \operatorname*{argmax}_{\delta\in(0,\infty)} \frac{h_\delta\delta}{\sqrt{f_{cc}^\lambda(-\delta)} + \sqrt{f_{cc}^\lambda(\delta)}}.$$

Appendix C Computation of Restricted cc-Optimal Designs

To simplify the notation, given the large number of formulas involved, we drop the subscript Γ from h_Γ, μ_Γ, and η_Γ.

To compute the restricted cc-optimal design given Γ, we consider the design described in Remark 4.3(ii): $\xi_{z,\omega} = \begin{Bmatrix} z & \Delta \\ \omega & 1-\omega \end{Bmatrix}$. Its classical and standardized information matrices are

$$\mathrm{I}(\xi_{z,\omega}) = \begin{pmatrix} \eta^2 t_{z,\omega} & u_{z,\omega} \\ u_{z,\omega} & \frac{v_{z,\omega}}{\eta^2} \end{pmatrix} \quad \text{and} \quad \tilde{I}(\xi_{z,\omega}) = \begin{pmatrix} d_1^2 t_{z,\omega} & u_{z,\omega} d_1 d_2 \\ u_{z,\omega} d_1 d_2 & v_{z,\omega} d_2^2 \end{pmatrix}, \tag{C5}$$

where $t_{z,\omega} = \omega h^2(z) + (1-\omega)h^2(\Delta)$, $u_{z,\omega} = -z\omega h^2(z) - \Delta(1-\omega)h^2(\Delta)$ and $v_{z,\omega} = z^2\omega h^2(z) + \Delta^2(1-\omega)h^2(\Delta)$.

C.1 Restricted A-Optimal Designs

The optimization problem for the A-optimal design in case (ii) in Section 4.3 is

$$\xi_A^\Delta = \operatorname*{argmax}_{z,\omega} \frac{tv-u^2}{d_1^2 t + d_2^2 v}(\xi_{z,\omega}) = \operatorname*{argmax}_{z,\omega} \frac{4(1-\omega)\omega\, h^2(z)\, (\Delta - z)^2}{[g_A(z) - g_A(\Delta)]\omega + g_A(\Delta)}.$$

where $g_A(z) = h^2(z)(d_1^2 + z^2 d_2^2)$. Following calculations analogous to those in Appendix B.2, the optimal design's parameters are

$$w^* = \frac{\sqrt{g_A(\Delta)}}{\sqrt{g_A(z)} + \sqrt{g_A(\Delta)}} \text{ and } z^* = \operatorname*{argmax}_{z\in(-\infty,\Delta)} \frac{h(z)\, (\Delta - z)}{\sqrt{g_A(\Delta)} + \sqrt{g_A(z)}}$$

C.2 Restricted cc-Optimal Designs

For a known λ, the restricted cc-optimal design is

$$\begin{aligned} \xi_{cc}^{\lambda,\Delta} &= \operatorname*{argmax}_{z,\omega} \frac{tv-u^2}{(1-\lambda)\sigma_1^2 t + \lambda\sigma_2^2 v}(\xi_{z,\omega}) = \operatorname*{argmax}_{z,\omega} \frac{t_{z,\omega}v_{z,\omega} - u_{z,\omega}^2}{(1-\lambda)d_1^2 t_{z,\omega} + \lambda d_2^2 v_{z,\omega}} \\ &= \operatorname*{argmax}_{z,\omega} \frac{4(1-\omega)\omega\, h_\Gamma^2(z)\, (\Delta - z)^2}{[g_{cc}^\lambda(z) - g_{cc}^\lambda(\Delta)]\omega + g_{cc}^\lambda(\Delta)}. \end{aligned}$$

where $g_{cc}^\lambda(z) = h^2(z)[(1-\lambda)d_1^2 + \lambda z^2 d_2^2]$. Following calculations analogous to those in the appendix B.2, the optimal parameters are

$$\omega^* = \frac{\sqrt{g_{cc}^\lambda(\Delta)}}{\sqrt{g_{cc}^\lambda(z^*)} + \sqrt{g_{cc}^\lambda(\Delta)}} \text{ and } z^* = \operatorname*{argmax}_{z\in(-\infty,\Delta)} \frac{h(z)(\Delta - z)}{\sqrt{g_{cc}^\lambda(\Delta)} + \sqrt{g_{cc}^\lambda(z)}}$$

Appendix D Symbol Description

$\mathcal{X}$	dose space
x	a dose in $\mathcal{X}$
y	binary response 1 or 0
F	Dose-response in $\mathcal{X}$
$\boldsymbol{\theta}$	parameters
Γ	fixed response rate
μ_Γ	dose with response rate Γ
η_Γ	slope of the dose-response function at μ_Γ
$\mathcal{Z}_\Gamma$	dose space with $\eta_\Gamma(x-\mu_\Gamma)$ parameterization
s	dose in $\mathcal{Z}_{.5}$
z	dose in $\mathcal{Z}_\Gamma$, $\Gamma \neq 0.5$
W	dose-response in $\mathcal{Z}_{.5}$
H	dose-response in $\mathcal{Z}_\Gamma$, $\Gamma \neq 0.5$
ν	μ_Γ in $\mathcal{Z}_{.5}$
ζ	slope ratio $\mu_\Gamma/\mu_{.5}$
Ξ	set of designs in $\mathcal{Z}_\Gamma$
ξ	a design
ξ^*	optimal design
$I(\xi)$	information matrix
$t_\Gamma(\xi)$	component (1, 1) of $I(\xi)$ in $\mathcal{Z}_\Gamma$
$u_\Gamma(\xi)$	component (1, 2) of $I(\xi)$ in $\mathcal{Z}_\Gamma$
$v_\Gamma(\xi)$	component (2, 2) of $I(\xi)$ in $\mathcal{Z}_\Gamma$
$\mathcal{C}_\Gamma(\xi)$	$(t_\Gamma,\, u_\Gamma,\, v_\Gamma)(\xi)$
$C_\Gamma(\mathcal{Z}_\Gamma)$	curve space
$\mathcal{CH}_\Gamma$	convex hull of $\mathcal{C}_\Gamma(\mathcal{Z}_\Gamma)$
$\mathcal{VB}$	Vertical Boundary
Φ, ϕ	optimality criterion on Ξ, on $\mathcal{C}_\Gamma(\mathcal{Z}_\Gamma)$
$\mathcal{RB}$	Right Boundary of $\mathcal{CH}_\Gamma$
$\widetilde{I}(\xi)$	standardized information matrix